Marvellous O. Ajala

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 4+ years of experience building scalable ML, Computer vision & NLP systems.

EXPERIENCE

SeamHealth | Abuja, Nigeria — *AI Engineer* March. 2026 - present

- Designed a resilient **multi-agent agentic LLM pipeline** that **transforms raw data into polished, multi-chart PDF reports:** handling the full lifecycle from data ingestion → multi-agent AI analysis → visualization rendering → PDF generation, with <**2 mins latency** and <**$0.5 cos**t per report.
- Built a **production-grade FastAPI microservice** with JWT auth, Redis-backed checkpointing with in-memory fallback for zero-downtime, Server-Sent Events for real-time progress streaming, docker compose deployment via GitHub Actions CI/CD and monitoring system with Prometheus metrics (30+ custom metrics: latency by step, token consumption by agent, error rates by type) and structured JSON logging, with alerting rules for P99 latency >5min and error rate >5%
- Implemented in industry standard practises (**automated PII redaction**, **specialized AI agents**, **recursive time-splitting analysis** mechanism that autonomously breaks large datasets into manageable slices to stay within LLM context windows while preserving temporal granularity, **automated quality gate** and **LLM-as-judge validation**, **concurrent LLM throughput parallel requests** via an observed semaphore with configurable limits, balancing cost vs. latency)
- Technologies: FastAPI, Python, OpenRouter, Redis, Prometheus, JWT Auth, SSE, Docker, GitHub Actions

KBeauty Nigeria | Lagos, Nigeria — *ML Engineer (contract)* Dec. 2024 - Dec. 2025

- Built an **end-to-end Dermatology ML pipeline** from raw data collection through production deployment: designed automated ETL (survey ingestion, image download, multi-label matrix construction), trained a 17-condition classifier, and deployed via Dockerized FastAPI + Nginx)
- **Implemented patient data anonymization (PII redaction) and person-level train/test splits** to ensure no subject leakage: mapped real names to random codes across filenames and spreadsheets before model training.
- Designed a **two-stage computer vision inference pipeline**: Google Derm Foundation (TensorFlow) for feature extraction + custom PyTorch MLP classifier head, with rembg (U^2-Net) background removal as preprocessing
- Technologies: Python, PyTorch, TensorFlow, FastAPI, Docker, Computer Vision

JuiceLife | Lagos, Nigeria — *Machine Learning Engineer* Sept. 2022 - Oct. 2023

- Architected and implemented an **end-to-end data pipeline** that **reduced data acquisition time by 80%** automating image preprocessing and processing fitness tracking data for 5k+ active users.
- Developed and deployed **v2 of computer vision-based workout tracking** system using Tensorflow and OpenCV, **improving** exercise recognition **accuracy by 10%** and **reducing user churn by 15%** in 2023.
- Engineered a **production monitoring system** using **MLflow** and **Prometheus**, **maintaining 99% uptime** across multiple deployed ML models.
- Led the development of **AI Coach V1**, a **recommendation agent using NLP (LangChain)** to personalize workout plans, increasing user retention by 20%.
- Technologies: Tensorflow, OpenCV, MLflow, Docker, FastAPI, LangChain, Prometheus and Grafana.

Recop.it (Defunct) | Lagos, Nigeria — *Machine Learning Engineer* May 2022 - Sept. 2022

- **Scraped and cleaned 100K+** product listings **using Python (BeautifulSoup, Scrapy)**, reducing **data preprocessing time by 60%**.
- Engineered **V1 recommendation system** with **collaborative filtering and content-based approaches** (Scikit-learn), boosting click-through rates by 25% for e-commerce users.
- Deployed models using Docker and FastAPI.

- Technologies: Scikit-learn, BeautifulSoup, Scrapy, Docker, FastAPI.

Dryad Network [Omdena] | Remote (Berlin, Germany) – Junior ML Engineer Sept. 2021 - Nov. 2021

- Collaborated on a **wildfire prediction model** using sensor data (**NumPy, Pandas, Tensorflow Lite**), **achieving 92% precision** in fire-risk classification.
- **Automated data transformation pipelines**, reducing feature engineering time by 40% for a team of 10 engineers.
- **Managed documentation** to ensure efficient project tracking and collaboration **using Sphinx and Mkdocs**.

EDUCATION

University of Lagos, Nigeria — *Bachelor of Pharmacy* 2016 - Aug. 2023

- Relevant Coursework: Probability and Statistics, Research Data Analysis, Intro. to Computer Science
- Activities: Google Developers Science Club (GDSC Unilag), Student Hackathons

SKILLS

Machine Learning: Scikit-learn, Pytorch, Tensorflow, Jax, OpenCV, NLP (Transformers, LangChain).
Data Science: Pandas, NumPy, SQL, Prefect, MLFlow, Prometheus
OS/Tools: Python, Julia, Golang, FastAPI, Docker, Linux (Ubuntu), AWS, Git, Github
Soft Skills: Leadership, Problem-Solving, Teamwork, Communication, Critical thinking.

PROJECTS

- **MRI-Based Hearing Problem Classifier** *(Python, PyTorch, OpenCV, pydicom, MLflow) -* Developed a 3D ConvNet binary classifier on DICOM MRI scans to detect hearing problems. Built a custom training pipeline with MLflow tracking, stratified splitting, class-balanced loss, and patient data anonymization.
- **Antimalarial Hit Discovery** (*Python, PyTorch, MLflow, GCNN, GAT, Chemprop, VAE, ChEMBL) -* Built an AI pipeline for novel antimalarial compound discovery combining virtual screening and de novo design. Trained multiple models (random forest, graph neural networks, transformers, variational autoencoders) on ChEMBL data with MLflow experiment tracking across the full discovery pipeline.
- **PharmAssis**t - **LLM Pharmacist Assistant** (*Python, LangChain, LLMs) -* Built an LLM-powered assistant providing medication second opinions including drug interaction checks and side effect analysis through natural language queries.
- **SOP-Reviewer** (*Langchain, FastAPI) -* A Large Language Model (LLM)-based assistant that provides detailed feedback to graduate students on their Statement of Purpose (SOP).
- **DL-Framework** (*Numpy) -* Implemented neural network framework from scratch using NumPy, supporting common architectures.

LEADERSHIP/MEMBERSHIP

- May 2024 - present: **Co-organizer** Data Science for Healthcare workshop (DS4H) at Deep Learning Indaba.
- Jan - Oct 2023: **ML Healthcare lead, CohereForAI,** organised reading group and knowledge sharing sessions.
- 2020- 2023: **Core committee Member, GDSC Unilag,** Led pioneer community created for medical students.

PUBLICATIONS & PREPRINTS

- **Ajala, M., et al.** "Application of Artificial Intelligence in the Classification of Microscopical Starch Images for Drug Formulation." Preprint, Deep Learning Indaba 2022. [arXiv Link]
- **Curri n, C. B., et al.** "A Framework for Grassroots Research Collaboration in Machine Learning and Global Health." MLGH Workshop, ICLR 2023. [OpenReview]
- **Ajala, M., et al.** "Machine Learning Based Screening and Identification of Antimalarial Compounds." Manuscript under preparation. 2025
- **Taliya W., et al.** "From Passive to Participatory: Healthcare Workshop Design for Continued Community Engagement" Helina, 2025